

Problem Set 4

Mathematics for Social Scientists

EXERCISE 1. Determine whether the following functions are convex or concave (if any at all) over the domain that they exist:

i. $f(x) = x^4$

ii. $g(x) = 8x + 26$

iii. $h(x) = \sqrt{x}$

iv. $p(x) = -x^{131894}$

v. $r(x) = e^x$

vi. $s(x) = \log(x)$

EXERCISE 2. For the following functions, determine **the interval(s)** over which the function is convex or concave (if at all), and identify any saddle (inflection) points:

i. $f(x) = \frac{1}{x}$

ii. $f(x) = \frac{x^{1-\gamma}}{1-\gamma}$ for $0 < \gamma < 1$

iii. $f(x) = xe^x$

iv. $f(x) = \sin(x)$

v. $f(x) = \lambda e^{-\lambda x}$ for $\lambda > 0$

Remark: This is the exponential distribution

vi. $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

Remarks: This is the standard normal distribution

vii. $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

Remarks: This is the general normal distribution

EXERCISE 3. Find the zeros and local maxima/minima of the following functions.

(Recall: the zeros of f are x such that $f(x) = 0$, whereas the local maxima/minima are x such that $f'(x) = 0$)

i. $g : [-1, 1] \rightarrow \mathbb{R} : x \mapsto x^2$

- ii. $f : (0, \infty) \rightarrow \mathbb{R} : x \mapsto x \ln(x)$
- iii. $f : (0, \infty) \rightarrow \mathbb{R} : x \mapsto x^2 \ln(x)$

EXERCISE 4. For the following functions f ,

- (a) differentiate f
- (b) find the local maxima and minima by solving $f'(x) = 0$
- (c) test whether these points are minima or maxima by finding $f''(x)$
- (d) calculate the value of f at these points
- (i) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 2x^3 - 9x^2 - 24x + 8$
- (ii) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 3x^4 - 4x^3$
- (iii) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 10xe^{-\frac{x^2}{2}}$
- (iv) $f : (0, \infty) \rightarrow \mathbb{R} : x \mapsto -10 \cdot \frac{\log(x)}{x^2}$
- (v) $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto (x^4 - x^2)^2$

EXERCISE 5. Consider the function $f(x) = (x^2 + 2x)e^{-x}$.

- (i) Determine for which values of x the value of the function is equal to zero.
- (ii) Calculate $f'(x)$ and find the extreme points of f . What is the maximum/the minimum of the function?
- (iii) Does the function have inflection points?
- (iv) Sketch the function and specify whether it is convex/concave (in sections).